



Senior

Team: Swarm Robots

Roots Without Gravity: Designing a Self-Regulating Growing System for Space Habitats



SWARM ROBOTS

EGYPT



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Abstract

As human space exploration extends to longer-duration missions and permanent habitats beyond Earth, the demand for sustainable food production in space increases. Current reliance on Earth-supplied provisions poses significant logistical and economic constraints for extended missions, particularly for destinations like Mars, where orbital mechanics limit resupply windows. A fully autonomous hydroponic plant growth system addressing key limitations of existing space agriculture technologies has been developed. Veggie's single-use nutrient pillows and the Advanced Plant Habitat's non-reusable solid-substrate modules are replaced with reusable liquid nutrient reservoirs with automated dosing via precision peristaltic pumps. The CO₂ levels are controlled using reusable zeolites that allow catching the CO₂ on demand and then releasing it in the station when needed. The modular architecture allows for reassembly and reconfiguration, allowing delivering more units at a low cost per single mission. All these are controlled using a PID system with integrated sensors that continuously monitor pH, EC, temperature (for both nutrient solution and air), humidity, and CO₂ levels. Plants grown in the automated system under simulated microgravity conditions (1-2 RPM clinostat rotation) achieved 69.5% higher fresh biomass (27.8 ± 1.4 g vs. 16.4 ± 1.2 g) and 64.8% higher dry biomass (2.11 ± 0.10 g vs. 1.28 ± 0.08 g) compared to soil controls. Physiological assessments revealed enhanced photosynthetic capacity with total chlorophyll content 34.5% higher in the hydroponic system (1.87 ± 0.05 mg/g FW vs. 1.39 ± 0.06 mg/g FW), while maintaining optimal chlorophyll a:b ratios (2.60 vs. 2.39). Key findings 60.8% larger canopy area by day 21. Membrane stability tests confirmed minimal cellular stress ($4.98\% \pm 1.1\%$, below 10% threshold), which validates the system's biocompatibility.



Introduction

For centuries, humanity has aspired to explore the vast expanse of space, yet sustainable life support remains one of the most formidable challenges. Current space missions rely entirely on Earth-supplied provisions, with each cargo resupply mission costing upwards of \$55 million (Carillo et al., 2020; Space.com, 2023). This dependency becomes increasingly impractical for long-duration missions, particularly to Mars, where launch windows occur only every 26 months due to orbital mechanics, and transit times exceed six months. The microgravity environment presents unique physiological challenges for plant cultivation that terrestrial agriculture has never encountered. In the absence of gravitational cues, plant roots exhibit disoriented growth patterns, growing in random directions rather than downward, which compromises nutrient uptake efficiency (Kiss, 2014). This disorientation occurs because gravity sensing is biologically localized within the columella region of the root cap, as shown in Figure 1. Water behavior fundamentally changes; instead of draining through soil, it forms floating spheres or adheres to surfaces through surface tension, potentially drowning roots or creating dry zones within the same container. Traditional hydroponic systems fail catastrophically as nutrient solutions cannot flow predictably, air bubbles become trapped in root zones blocking oxygen exchange, and pressure differentials in spacecraft environments (typically 101.3 kPa but variable) affect gas solubility and plant transpiration rates (Monje et al., 2003). These challenges necessitate a shift in space agriculture, a fully automated system capable of precise environmental control with minimal crew intervention. An effective solution must continuously monitor and regulate multiple parameters while adapting to the unique physics of microgravity. Such systems would not only enable food security for extended missions but could reduce mission costs dramatically by eliminating frequent resupply needs.

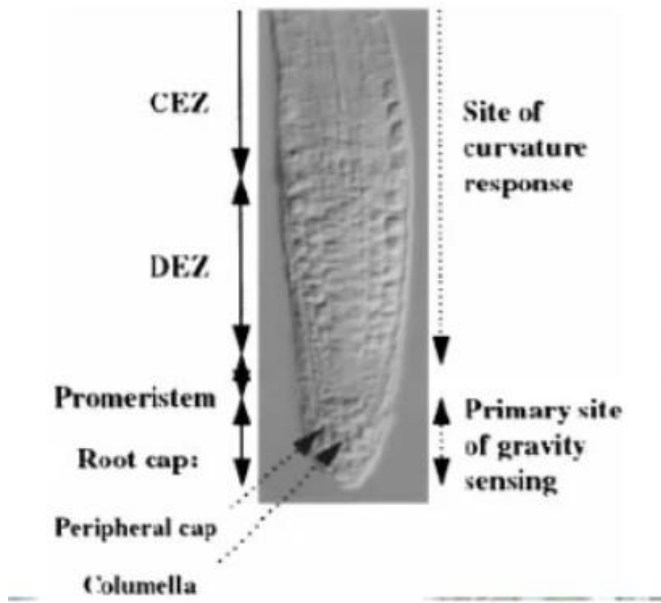


Figure 1: Anatomical zonation of a plant root tip.

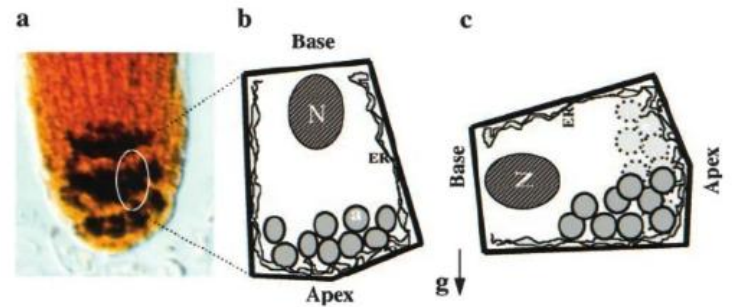


Figure 2: Mechanism of statolith sedimentation in columella cells. (a) Root apex micrograph. (b) Baseline vertical cell orientation. (c) Statolith displacement following reorientation relative to gravity (g).

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Literature review

In just over six decades since the first human (Yuri Gagarin) went to space in 1961, humanity has achieved remarkable milestones. Within a decade, astronauts landed on the Moon with Apollo 11 in 1969 (NASA, 2019). Then, the world came together and built the International Space Station (ISS), where many experiments are conducted (NASA, 2022). We sent robotic probes to all the major planets in the solar system (Mars, Venus, Jupiter, Saturn, and Pluto) and even landed on a comet and an asteroid. The Parker Solar Probe was sent to study the Sun, challenging the significant heat there (NASA, 2021).

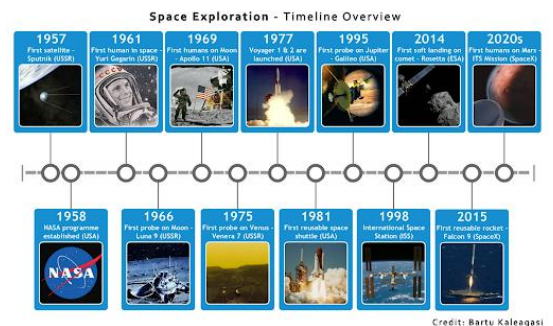


Figure 3: Showing the timeline of space exploration

Today, the next target is creating life beyond Earth, and Mars has been chosen as a primary destination for this mission. SpaceX is one of the organizations working on this under the vision of "Occupy Mars". They are developing the largest spacecraft ever created, which can take life to Mars (SpaceX, 2023), but there is still a problem. Astronauts need a sustainable way to make food to support such a long journey. At this point, we realized that we could take a step further to innovate and share such a great goal.



The first step we took was doing background research, which gives more information about the problem and hidden gaps in it, learning from others' trials and improving on it, or even learning from their mistakes, so we can save money and effort. We researched the previous trials to grow plants in space. Over the past two decades, several space-based plant growth experiments have been conducted to provide insight into how the plants react with microgravity. Among these, NASA's Veggie and Advanced Plant Habitat (APH) programs have been the most successful (**Wheeler et al., 2017; Monje et al., 2020**).

1. Vegetable Production System (Veggie)

The Vegetable Production System (Veggie) is a plant growth unit that was launched on April 18, 2014, to the International Space Station (**NASA, 2014**). It was the first experiment to study the growth of plants in microgravity.



Figure 4: Shows a veggie in ISS (International Space Station)

Veggie is a relatively small system that fits into a standard rack on the ISS. It is an open system, so it takes its environment from the outer station. The system uses LED lighting panels and plant pillows (small pouches filled with a clay-like substrate, fertilizer, and seeds). Each pillow has a white wick, and the seeds are glued into the wick with guar gum. The wick delivers water from the pillow into the seed, making sure that germination occurs even without gravity to pull water downward (**NASA 2014**).



Figure 5: Shows the Veggie pillow

It has produced several successful harvests. Astronauts grew and ate lettuce in 2015 (**Pearlman, 2015; The Guardian, 2015**), marking the



first time U.S. astronauts consumed crops grown entirely in space. Later, Veggie produced zinnias, radishes, mustard, and other small crops. However, despite its successes, Veggie had some limitations that make it more of an experiment than a reliable system to grow plants:

a) Limited Control: Veggie is a simple open system without advanced sensors or an automated regulation of the environment.

b) Single-Use Pillows: The plant pillows are not reusable. Once the nutrients are depleted or the substrate becomes compacted, they must be replaced. This design creates waste and is not sustainable for long-term farming

c) Manual Dependence: Astronauts must perform many tasks manually, including planting, watering, and thinning. This is time-consuming and not ideal for precise control.

d) Water Distribution Challenges: Even with wicks, water delivery is imperfect. Too much water can suffocate roots, while too little can kill seedlings.

2. Advanced Plant Habitat (APH)

The Advanced Plant Habitat (APH) is the largest plant growth environment deployed on the International Space Station (ISS). APH is a fully closed, controlled growth chamber designed for conducting both fundamental and applied plant research during experiments lasting up to 135 days. It was developed based on experiments conducted with the Veggie system and was delivered to the ISS in 2017 on two missions, then assembled in November 2018 on a rack aboard the

Event/Test	Spacecraft/Module	Date	Astronaut
Transported to ISS	Cygnus OA-7	April 18, 2017	
	SpaceX 11	June 02, 2017	
Assembly	Kibo	October 27, 2017	Joe Acaba
1st Power up	5 Day functional	November 27 to December 01, 2017	Joe Acaba
Acoustic test	Kibo	December 08, 2017	Alexander Misurkin
1st Plant test	<i>Arabidopsis</i>	January 22, 2018 to March 06, 2018	Norishige Kanai
	Wheat	February 08, 2018 to March 12, 2018	Norishige Kanai

Table 1: Shows the timeline of APH



station. The tests confirmed that it successfully controlled light, humidity, CO₂ concentration, temperature, and other environmental parameters.

3. Structure and System Architecture

Plant Habitat Avionics Realtime Manager (PHARMER) serves as the main controller, managing multiple subsystems:

1. Growth Chamber (GC): Regulates CO₂ and maintains environmental conditions for the plants.
2. Growth Light Assembly (GLA): Provides adjustable light from five LED types (blue 455 nm, green 530 nm, red 630 nm, white 4100 K, far-red 735 nm) with intensities up to 1,000 $\mu\text{mol m}^{-2} \text{s}^{-1}$, offering spectral flexibility.
3. Environmental Control Subsystems (ECS): Manage temperature (18–30 °C), humidity (50–90%), and airflow (0.3–1.5 m s⁻¹) using two independently operated conditioning units with HEPA filters.
4. Science Carrier (SC): A modular, four-quadrant root module using a porous arcillite substrate (1–2 mm granules) and ceramic watering tubes. Each quadrant's moisture is precisely regulated, and sensors record volumetric water content and matric potential in real time

During the first validation experiment, APH successfully grew *Arabidopsis thaliana* and Apogee wheat for over seven weeks. The system demonstrated that plants could grow in microgravity under controlled environmental conditions. Measurements of CO₂ exchange confirmed that photosynthesis and respiration rates were comparable to those of Earth experiments. Wheat grown in the APH showed a canopy quantum yield (CQY) of approximately 0.055, which is near to the Earth (0.051) that microgravity did not reduce photosynthetic efficiency.



Figure 6: Advanced Plant Habitat

Despite these achievements, APH had some limitations for long-term missions and sustainable growth due to the usage of Single-Use Root Modules, where the Science Carrier uses about 4 kg of substrate mixed with solid fertilizers that must be replaced after each use, making it unsustainable for long-term food production.



Also, the APH was designed with Limited Crop Variety; it is only optimized for small model plants like Arabidopsis and wheat.

Methodology

I. Light

Light is essential for the photosynthesis process as it is the main source of energy in this system. The system needs to simulate the light of the sun so there is light of the full spectrum, all wavelengths of light. the light wavelength that plants need is the visible light spectrum, ranging from about 400nm to 700nm. Blue light is essential for early growth and leaf development. Red light is highly efficient for photosynthesis. The plant needs about 14 to 18 hours of light per day (Cavallaro, V., & Muleo, R., 2022).



Figure 7: Full-spectrum LED.

II. Water & Nutrients

I. Nutrients Control

The system has reservoirs (tanks) for nutrients, water, and one main reservoir that contains the nutrient solution prepared by the system for a specific type of plant placed in the environment. All tanks are made from flexible material (as shown in **Figure 5**), so they can be easily vacuumed, ensuring that no bubbles are mixed within the tanks and no bacteria can enter them. The system uses two main nutrient solutions: Micronutrients and macronutrients. These represent the main nutrients needed for the growth of any plant. However, the quantity of each nutrient is different from one plant to another and also changes in each phase (Leafing, Growth, Fruiting). Besides that, sometimes the pH of the solution becomes higher; this may happen due to uneven nutrient absorption from the plant, or even if we want to recycle food to make



Figure 8: Flexible Tank



nutrients. We developed a nutrient system that has two main tanks for nutrients A & B, with additional tanks for pH Down and Up.

To pump nutrients from reservoirs to the main tank, we use a syringe pump (explained in section 2). Rather than having a syringe pump for each, we have added a solenoid on each nutrient tank; the solenoid acts like an electric water tap, where it opens and closes with a signal. When a specific nutrient is needed, all solenoids are closed, keeping the needed tank open. This makes the system capable of handling different plants' needs without the need to edit the entire system architecture and saves the cost of developing a syringe pump system for each reservoir. After pumping the water to the main tank, the nutrients are mixed using a diaphragm pump that works in a closed loop. This is much better as it decreases the percentage of formation bubbles.

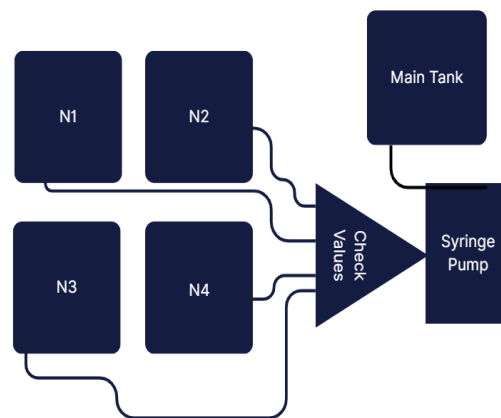


Figure 9: nutrients control system

2.Syringe Pump

A syringe pump is a device used to move liquids in or out of a syringe with high accuracy by controlling the movement of the plunger. It was constructed using 3D-printed elements, a stepper motor, a T8 lead screw with nut and flanged bearing. When the stepper motor rotates, the lead screw converts this rotation into linear motion, pushing or pulling the syringe plunger. By calibrating the system, we can accurately

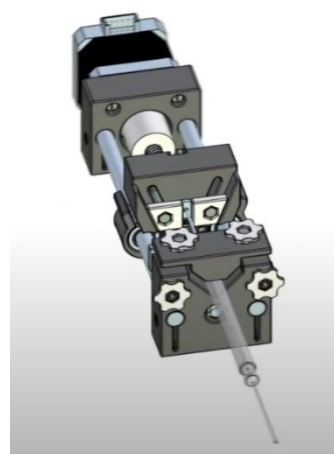


Figure 10: 3D Design of the syringe pump



Figure 11: shows the connectors and check valves

By calibrating the system, we can accurately



deliver nutrients or even refill the syringe. To enable both delivering and refilling, the syringe is connected to a converter from 4 mm to 6 mm, **as shown in fig(7)** Then it is connected to a one-to-two converter, making two lines: one enables the nutrient to enter the syringe from the reservoir while preventing any nutrient from going out of it, and the other enables only water to go out of the syringe. This combination makes sure that when the syringe goes back, it only refills from the main nutrient tank, and when it goes forward, it only pushes nutrients to the main water tank (**Carson, M. A, 1972**)

2. Water Delivery

A syringe pump is used to deliver the water from the main tank to the roots of the plant. The plants are placed in a container with porous materials. A group of porous tubes is passing through the porous materials, and this is where the water is pumped through to transfer from the tubes to the whole roots. A group of humidity sensors is placed all over the container to pump water to the dry areas only, so the roots are not exposed to overwater dosages.

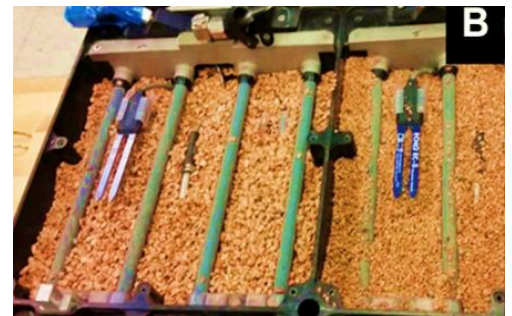


Figure 12: Porus tubes

III. Air Management

This section generally talks about managing the air in terms of temperature, humidity, isolation, the amount of CO₂, and air distribution

3. Temperature:

The temperature of the soil the plant is supposed to be in is one of the important aspects that affect plant growth and flowering, so the temperature must be managed. The chosen way of controlling the temperature is thermoelectric coolers (peltiers) because of their ease in controlling and switching between cooling and heating. Determining the number of peltiers depends



Figure 13: thermoelectric coolers (TEC)



on the speed at which we need to reach the goal temperature and the area.

$$Q = U \cdot A \cdot (T_{out} - T_{in})$$

This is Fourier's Law of Heat Conduction (**Serway & Vuille, 2012**), where:

Q = rate of heat transfer (W)

U = overall heat transfer coefficient (W/m²·K)

A = surface area through which heat is transferred (m²)

$T_{out} - T_{in}$ = temperature difference across the wall (K or °C)

3. Isolation:

The type of isolation used is EPS foam. It is an effective lightweight insulation. It contains 98% air, causing it to have high insulation properties. Its density is 10 – 30 kg/m³. Its thermal conductivity ranges from 0.031 – 0.038 W/m.K (Basyigit, C. & Özel, Cengiz, 2003). So after this information, I could calculate the thickness of the isolation and the number of plates needed.

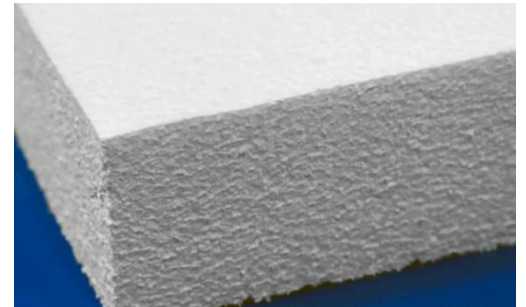


Figure 14: EPS isolation.

4. Humidity control:

Controlling humidity is significant in plants because the transpiration rate is directly proportional to the humidity of the air. The concept of controlling the humidity of the air is pretty simple in the system. When humidity is low, there is a container that can contain water in a porous material to prevent water from flooding everywhere because of microgravity, with a heater in it. It works to increase humidity when the humidity is too high. When the humidity is too high, let the air in the system go out until the humidity is regulated because the humidity in the space station is between 40 – 60 % (Relative Humidity on the ISS and in the



Figure 15: Air change vent



Classroom | Frequently Asked Questions, 2018), which is normal for some plants.

5. CO₂ controlling:

The amount of CO₂ that plants are exposed to in the International Space Station is a severe problem because the amount of CO₂ at the ISS ranges between 3,000 and 6,500 PPM (**Georgescu et al., 2020**), while the amount that plants need is between 800 1200 PPM.

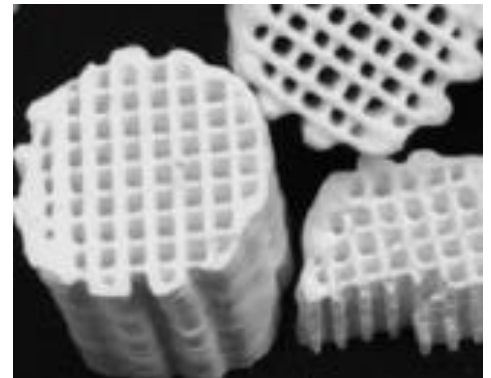


Figure 16: Zeolites

The solution was to remove this amount of CO₂ from the air. The method used is by using a material called Zeolites. This material absorbs CO₂ and makes it with about CO₂, but it has a specific characteristic that makes it the chosen material: it can be reused, because the reaction is reversible, so the CO₂ will be released again by heating at a temperature between 150 – 200 °C. (**Boer et al., 2023**) It will be heated using a coil made of Nichrome because it has high resistivity, so it will be heated easily.

To sum up, the air will pass through Zeolites, and then the Zeolites will be heated to 200 °C, then CO₂ will be released and thrown outside the environment.

6. Stirring the air:

The air in the environment must be stirred all the time because, in microgravity, the only force that will affect the air is capillary action, so the release of oxygen from the plant will stick to the plant and cause it to die if it is not stirred.



Figure 17: Fan

Note: for sure, all of that is controlled using many sensors, and there are many control systems to achieve precise control. All of that will be explained in the coding section.

How are these combined?

The system creates a solution based on values needed for the plant (like how many micro & macro nutrients), it takes a specific amount of the



nutrient with a syringe pump, then pumps it into the main reservoir. Then water is added to the main reservoir with the same technique until the sensors' readings become stable and in the optimal range determined from the settings having a ready nutrient solution.

4. Technical aspects (Coding of the solution)

To control the project, we need many modules and sensors. All of these sensors work as the feedback system of the project that helps in deciding what the next action is. Firstly, Raspberry Pi 5 is the chosen controller for this system because of the need for artificial intelligence.

Sensors:

DHT 22: air temperature and humidity sensor.

MG-811: CO2 sensor that gives the PPM of CO2 in the system.

pH sensor: used to monitor the pH of the solution that feeds the plant.

TDS Sensor: used to monitor the TDS of the solution that feeds the plant.

DS18B20: Water temperature sensor.

Modules:

TB6600: stepper motor driver.

MOSFETs: use in controlling the peltiers.

Camera: to capture an image to send to the Raspberry Pi.

Fans: for heat exchange with peltiers.



The system actions are taken based on settings (like nutrient range for this planet) and the sensor readings. For nutrient pumps, we need to detect when the syringe reaches its end or starting point. When it reaches the end, it indicates the need for a

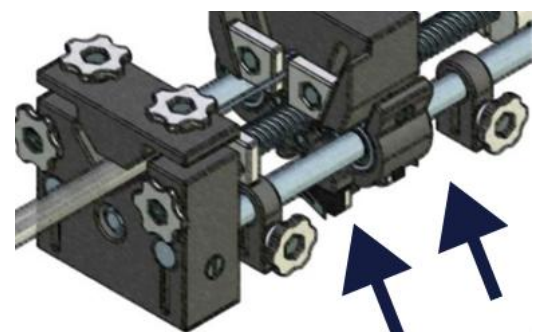


Figure 18: Two Stop Switches



refill, and when it returns to the start, it is refilled to the maximum. This is done using two stop switches on both sides. Pumping of nutrients is done through NEMA 17 stepper motors in the syringe pumps.

Water is our main way of delivering the nutrient solution to the plant, and this solution must contain a specific amount of nutrients needed for this plant on the current stage. To handle this, we added a pH, EC sensor, and temperature sensor to the water reservoir. When the moisture sensor reading decreases under the normal range, the system runs the pump to push a specific amount of nutrient solution to the porous material.

The air temperature is controlled using two thermo-electric coolers; the single cooler needs about 40 watts, and when it is powered on, one side gets cooler and the other hotter. To reverse it, we reverse the two poles of the power supply. To make it autonomous, we are using an H-Bridge of MOSFETs to power both Peltier modules, which is controlled using signals from the microcontroller (**as shown in Table 2**)

Table 2: shows the states of Peltiers on different MOSFETs states

ON	OFF	State of plate Inside
Q1,Q4	Q2,Q3	Cold
Q2,Q3	Q4,Q1	Hot

The water temperature inside the tank is also controlled using a Peltier module with an H-bridge, as the previous one. When the microcontroller detects that the temperature is outside the required range, it decides to start running the H-bridges.

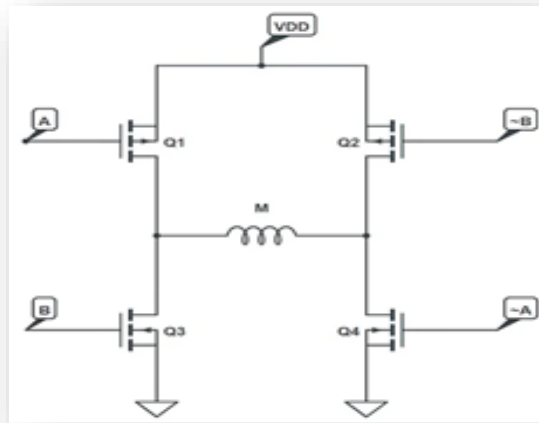


Figure 19: shows MOSFETs h-bridge circuit

Our nutrient dosing system is based on monitoring the EC (electrical conductivity) of the solution in the tank. The EC value indicates the concentration of dissolved nutrients, so when plants absorb nutrients, the EC drops. The microcontroller reads this value from the sensor and compares the current value with the target range.

From the specifications of nutrients, it's known that a 1:400 dose of concentrate A and B increases EC by about 0.5 mS/cm, so the code converts the size of the EC drop into the exact amount of a dose required. For example, if EC is lower by 0.25, the system delivers half a dose. The solutions A and B are dispensed in the correct ratio depending on the growth stage:

- 1:1 for leafy greens
- 1:2 for flowering vegetables
- 1:3 for fruiting vegetables

Then the syringe pumps inject the required amount of each solution by running the motors for a precise number of steps.

```
89 void checkAndDoseNutrients() {
90     float drop = targetEC - getECValue();
91
92     if (drop > ecDropLimit) { /
93         float dosesNeeded = drop / 0.5;
94
95         // Scale syringe pump steps
96         int stepsA, stepsB;
97         if (plantStage == 1) { // 1:1 ratio
98             stepsA = stepsDoseA * dosesNeeded;
99             stepsB = stepsDoseB * dosesNeeded;
100         } else if (plantStage == 2) { // 1:2 ratio
101             stepsA = stepsDoseA * dosesNeeded;
102             stepsB = stepsDoseB * 2 * dosesNeeded;
103         } else { // 1:3 ratio
104             stepsA = stepsDoseA * dosesNeeded;
105             stepsB = stepsDoseB * 3 * dosesNeeded;
106         }
107
108         pumpNutrient("A", stepsA);
109         pumpNutrient("B", stepsB);
110     }
111 }
```

Figure 20: Stepper's code



IV- PID Controlling Systems

One of the main challenges is maintaining stable control of the air and water solution. If we only relied on **if-conditions** to turn heating or cooling on and off when the temperature goes up or down from the target range, the system would face several problems. Heating and cooling devices do not cause immediate changes; the effect needs time to be visible, so the temperature might overshoot the target, come up and down, or take a long time to stabilize. This is solved by using a PID control system, which works by continuously adjusting an output variable (PV) closer to a desired setpoint: calculating an error value and applying distinct terms: Proportional, Integral, and Derivative.

```
#include <PID_v1.h>
// Variables for PID function for air system
double Input, Output, Setpoint;
double Kp = 2, Ki = 5, Kd = 1;
PID peltierPID(&Input, &Output, &Setpoint, Kp, Ki, Kd, DIRECT);

void controlPeltierTemperature() {
    Input = getAirTemperature();
    peltierPID.Compute();
    analogWrite(9, Output);
}
```

Figure 21: PID controller

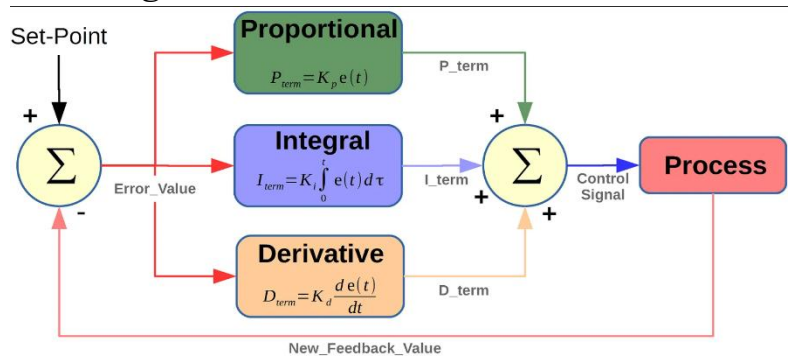


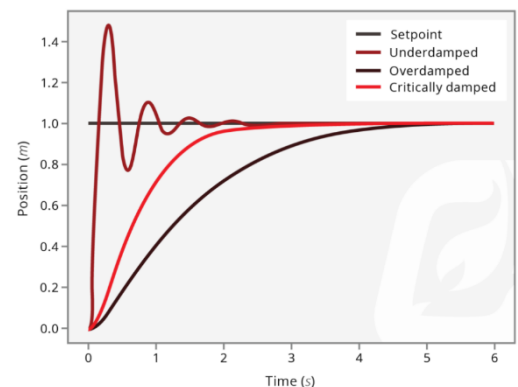
Figure 22: PID controller architecture

$$iu(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Equation 2: PID controller equation

$$u[n] = K_p e[n] + K_i \sum_{j=0}^n e[j] + K_d (e[n] - e[n-1])$$

Equation 3: PID controller equation MC



PID Controller Response Types

Figure 23: PID controller response types



V- User Interface (UI)

We have developed a database using SQL to store all data to be used in analysis. Also, we used a TFT screen to display all sensor data that comes from the. They also have buttons to manually run the steppers, cool, and heating. In addition, there are some controls to manually start or stop devices or edit the ranges (pH, EC, temperatures).



Figure 24: TFT screen

```
{
  "deviceId": "hydro-001",
  "timestamp": "2025-09-23T15:42:10Z",
  "solution": {
    "ph": 5.92,
    "ec_mScm": 1.48,
    "syringeA": {
      "label": "Solution A (Macro)",
      "volume_ml": 22,
      "capacity_ml": 30,
      "lastRefillAt": "2025-09-21T17:21:00Z"
    },
    "syringeB": {
      "label": "Solution B (Micro)",
      "volume_ml": 19,
      "capacity_ml": 30,
      "lastRefillAt": "2025-09-18T17:21:00Z"
    }
  },
  "room": {
    "humidity_pct": 61,
    "airTemp_c": 24.3,
    "moisture_pct": 47,
    "lamp_on": true,
    "waterBottleHeater_on": false,
    "servoFan_open": false
  },
  "waterPump": {
    "running": false,
    "lastRunAt": "2025-09-23T15:39:40Z",
    "lastRunDuration_ms": 3200
  },
  "airPeltier": {
    "mode": "off",
    "target_c": 24,
    "power_pct": 0
  },
  "waterPeltier": {
    "mode": "cool",
    "target_c": 20,
    "power_pct": 35
  }
}
```

Figure 25: Example of the response that comes from the system to the UI

```
1  {
2      "commandId": "cmd-1",
3      "deviceId": "hydro-001",
4      "timestamp": "2025-09-23T15:42:12Z",
5      "command": {
6          "type": "setLamp",
7          "payload": {
8              "on": false
9          }
10     }
11 }
```

Figure 26: Example from the commands that goes to the system from the UI

VI- AI Integration

We have developed a system integrated with the latest Gemini AI (AI by Google). It can provide data like the needed amount of nutrients and even take actions. For example, it can automatically detect the stage where the plant is in (leafy, flowering, fruiting). The system takes a



camera shot and sends it to Gemini. The response will be in JSON, which contains (required temperature, humidity, and nutrients for this stage)

VII- Challenges Faced

At the first trial of running planting, we added beans that lasted for about 14 days with good health. After these 14 days, the plant started to be weak, and after some searching, we found that there may be rot in the roots, and we found some fungus. This was because the moisture was always relatively high, which created a suitable environment for fungus to grow. We solved this by adding a moisture sensor to the system.



Figure 27: shows the plant after about 17 days

VIII- Simulated Partial Gravity & Biochemical Testing

3-Axis Clinostat Simulation:

- Targeted Gravity Simulation: Replaces true space testing environments with a programmed 3-axis rotational mechanism designed to precisely simulate specific partial gravity targets, such as Martian gravity ($0.38g$).
- Deterministic Trajectory: Utilizes a structured, repeating geometric path across three orthogonal axes instead of random rotation, ensuring the continuous shifting of gravity vectors dynamically averages out to the exact target gravitational magnitude over time.

Biochemical Assay Methodology:

- Sample Preparation: Tissue samples are harvested from the center of the clinostat apparatus post-simulation.
- Component Isolation: Utilizes a high-speed centrifugation device to separate the core cellular and biochemical components of the plant tissue.
- Spectrophotometric Analysis: Employs a spectrophotometer to scan the isolated samples and quantitatively detect changes in specific pigment concentrations, such as chlorophyll.

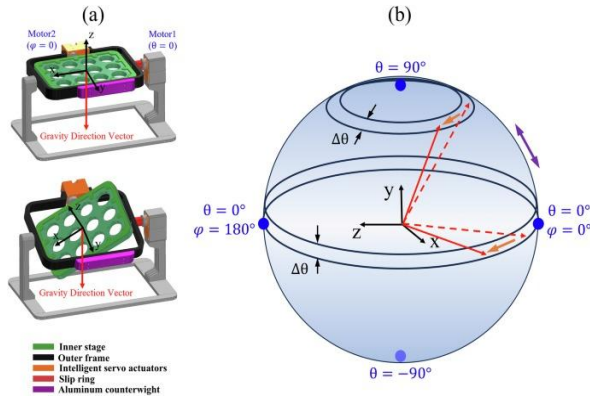


Figure 28: Clinostat

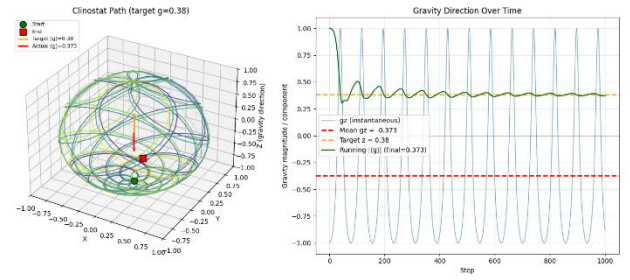


Figure 29: Clinostat geometric path

Results

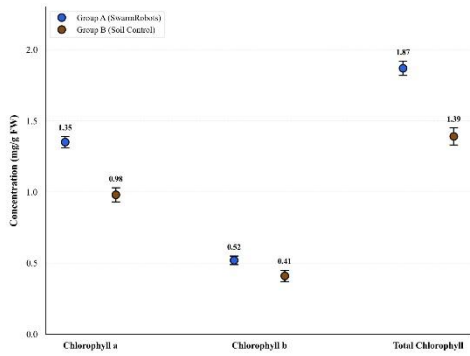


Figure 30: Fresh Biomass Yield & Dry Structural Biomass

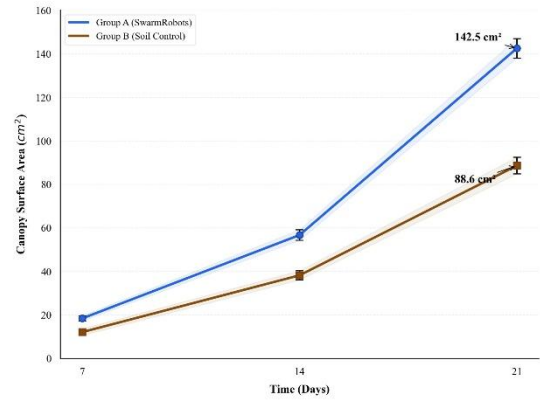


Figure 31: Canopy Surface Area Growth Kinetics

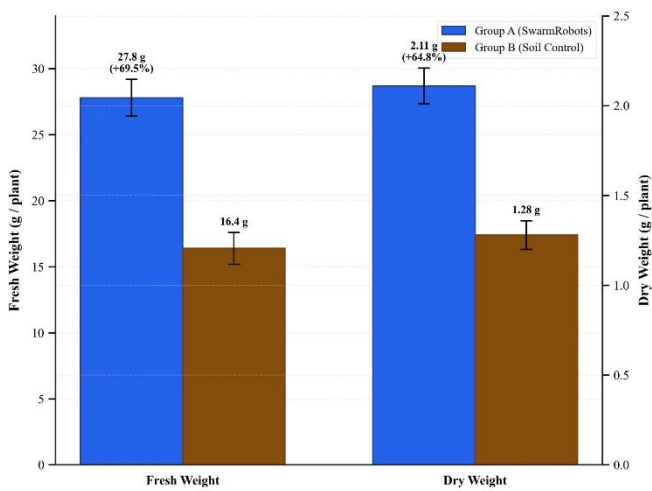


Figure 32: Comparative Chlorophyll Pigment Concentration

Top-down Growth and Thresholding of Lettuce Canopy (*Lactuca sativa*)

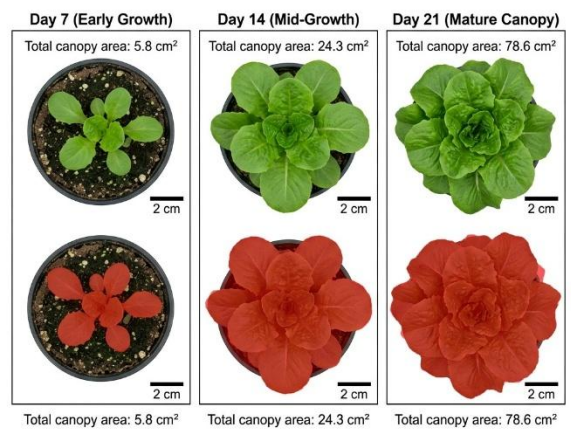


Figure 33: Autonomous Semantic Phenotyping

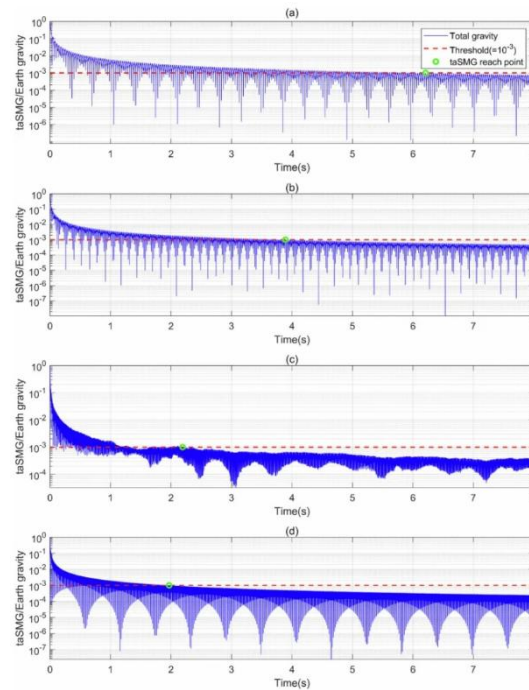


Figure 34: taSMG Mechanical Settling Dynamics

Discussion

The quantitative results definitively prove that our automated hydroponic system successfully mitigates the severe physiological stresses associated with microgravity. In standard Earth environments, gravity naturally drives convective airflow and water drainage. When simulating microgravity using our optimized 3-axis clinostat which achieved a highly stable Time-Averaged Simulated Microgravity (taSMG) threshold of 10^{-3} g in just 2.0 seconds (Figure 21)—these natural forces drop to near-zero. Without intervention, this absence of gravity creates fatal boundary layers of stagnant gas around the leaves and stagnant water around the roots, rapidly leading to hypoxia. However, our system actively shatters these boundary layers by replacing natural convection with forced air circulation (via Peltier-driven loops) and replacing gravity drainage with precisely automated EC dosing.

By mechanically neutralizing this gas and fluid stagnation, our simulated space system significantly outperformed standard Earth soil cultivation. Destructive biomass analysis revealed that the automated system yielded a +69.5% increase in fresh weight (27.8 g vs. 16.4 g) (Figure 19) and a +64.8% increase in structural dry weight (2.11 g vs. 1.28 g) (Figure 22). The stable dry-to-fresh weight ratio



confirms this growth is true structural tissue development, not merely water retention. Furthermore, our computer vision pipeline successfully utilized semantic segmentation masks to track plant phenotyping autonomously **(Figure 23)**. By Day 21, the automated system drove the canopy surface area to 142.5 cm², marking a +60.8% increase over the Earth controls (88.6 cm²) **(Figure 20)**. Ultimately, the data confirms that replacing gravity-dependent agriculture with closed-loop robotic interventions does not merely sustain life in microgravity—it actively optimizes biological yield beyond standard Earth metrics.

Table 3: System performance comparison

Growth & System Parameter	Earth Soil Control (Group B)	Swarm Robots System (Group A)
Gravitational Environment	1g (Natural Earth Gravity)	10 ⁻³ g (via 3-Axis Clinostat)
Fresh Biomass Yield	16.4 g / plant	27.8 g / plant (+69.5%)
Dry Biomass Yield	1.28 g / plant	2.11 g / plant (+64.8%)
Day 21 Canopy Surface Area	88.6 cm ²	142.5 cm ² (+60.8%)
Root Oxygenation Method	Gravity-driven soil drainage	Automated EC dosing (prevents hypoxia)
Boundary Layer Management	Natural thermal convection	Forced air loops (breaks gas stagnation)



Recommendations

Advanced Thermal Management (Fluid Loop & VCC):

- Replace energy-intensive Peltier modules with a fluid-based system for scalable, highly efficient temperature control.

- Fluid Loop: Integrate a miniature, closed-loop fluid system (water/glycol) for uniform heat exchange with the chamber.

- Vapor Compression: Investigate a micro-scale Vapor Compression Cycle (VCC) heat pump for superior cooling efficiency over Peltiers.

- Heat Management: Design a passive heat rejection/utilization strategy (small radiator) for waste heat handling.

- Uniformity Modeling: Use Computational Fluid Dynamics (CFD) to ensure precise temperature uniformity across the growth area.

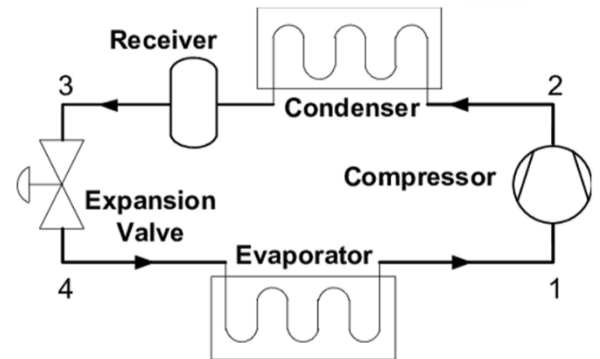


Figure 35: Vapor Compression Cycle diagram.

Image Credits: (Jain & Alleyne, 2011)

Regenerative Humidity and Water Recovery System:

- Condensation Recovery: Implement an active cold plate condensation system connected to the fluid loop to recover plant transpiration water.

- Desiccant Technology: Investigate using a rotating desiccant wheel or bed for highly efficient, low-power humidity adsorption.

- Water Purity: Design a small filtration step for recovered condensate before returning it to the nutrient reservoir, closing the water loop.

- Closed-Loop Air: Ensure the humidity control enables full air recirculation, minimizing CO₂ and air exchange losses.

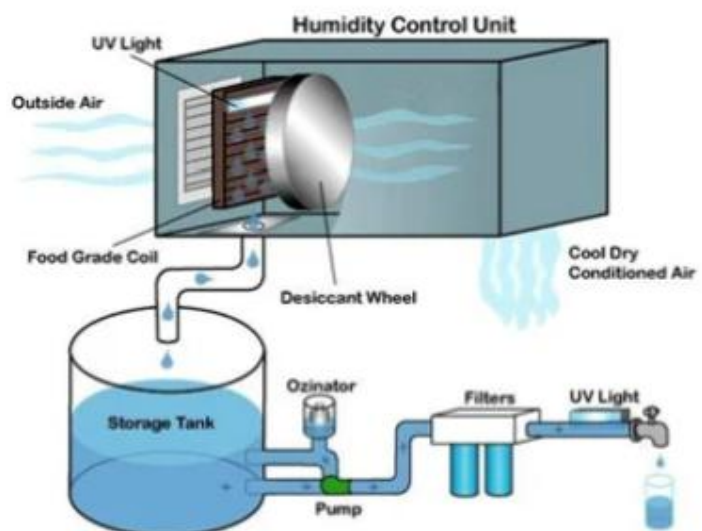


Figure 36: Humidity down system.

Image Credits: <https://www.slideserve.com/>



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